Replicating Huggett (1996): Wealth Distribution in Life–Cycle Economies

Quantitative Macroeconomics & Numerical Methods

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March 27, 2026

Abstract

The project replicates the main results of Huggett (1996), who studies the determinants of wealth inequality in an overlapping–generations economy with idiosyncratic earnings shocks, uncertain lifetimes, and borrowing constraints. The household problem is solved using the Endogenous Grid Method (EGM) of Carroll (2006) and the stationary wealth distribution is computed via Monte Carlo simulation of 500,000 agents. The replication closely matches the paper's baseline moments for the uncertain–lifetimes specification with earnings shocks and borrowing limit $a' \geq -w$: $\frac{K}{Y} = 3.323$ (paper: 3.2), wealth Gini coefficient $= 0.746$ (paper: 0.76), top 1% wealth share $= 11.8\%$ (paper: 11.8%), and top 5% wealth share $= 35.4\%$ (paper: 35.6%). This paper discusses results, computational choices made in the replication, how they differ from Huggett's original numerical approach, and the implications of those differences for accuracy and efficiency.

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1 Introduction

Understanding the sources of wealth inequality has always been a fundamental question in economics. Data from the 1983 Survey of Consumer Finances (SCF) show that the wealth distribution in the U.S. is substantially more concentrated than that of earnings. Specifically, while the Gini coefficient for earnings is approximately 0.63, the corresponding figure for wealth exceeds 0.76 (Huggett, 1996). Moreover, the top 1% of households hold nearly 30% of total wealth, and roughly one–quarter of households have zero or negative net worth.

Huggett (1996) investigates whether a calibrated overlapping–generations (OLG) model with idiosyncratic earnings shocks can account for these empirical findings. To do so, the author builds a model where agents live from the age of 20 up to 98, face persistent stochastic earnings shocks, and make consumption–savings decisions subject to borrowing constraints. The intuition is that the interaction of life–cycle savings, precautionary savings, and borrowing constraints can generate a wealth distribution that is substantially more concentrated than the underlying earnings distribution.

This project presents a computational replication of Huggett (1996) using modern numerical methods, studied throughout the course. The household problem is solved with the Endogenous Grid Method (EGM) by Carroll (2006), which inverts the Euler equation and thereby avoids non–linear root finding at each grid point. As discussed in Goensch (2025), EGM can deliver speed gains of up to 30$\times$ relative to value function iteration in life–cycle problems of this kind. The stationary wealth distribution is then computed by Monte Carlo simulation of a large panel of agents using inverse CDF sampling. General equilibrium is obtained by iterating on aggregate capital and transfers until the market–clearing conditions are satisfied, using a damped updating procedure. The age–earnings profile, which Huggett presents only graphically in Figure 1, is taken from the VFI Toolkit replication of Kirkby (2022). No official replication files or datasets were available online. Overall, the replication closely matches the original results for the uncertain–lifetimes economies, while the certain–lifetimes display a systematic upward bias in the capital–output ratio.

2 Model

The model represents a stationary overlapping–generations economy with idiosyncratic earnings uncertainty, uncertain lifetimes, and a social security system.

2.1 Demographics

Time is discrete, and the economy is populated by overlapping generations of agents. A new cohort enters the economy at age $t = 1$ corresponding to real age 20. Agents may live up to $N = 79$ periods, that is, age 98. Retirement occurs at age 65, corresponding to $R = 46$.

In the uncertain–lifetimes specification, the conditional probability of surviving from age $t-1$ to age t is denoted by $s_t \in (0, 1]$. Agents live for at most N periods, so death is certain after the terminal age. Survival probabilities are calibrated using the actuarial estimates in Jordan (1975). With population growth at the constant rate $n > 0$, the stationary mass of age t agents satisfies:

$$\mu_t = \frac{\prod_{j=2}^{t} s_j}{(1+n)^{t-1}} \mu_1 \quad \text{for} \quad t = 1, \ldots, N \quad \text{with} \sum_{t=1}^{N} \mu_t = 1 \tag{1}$$

In the certain–lifetimes specification, survival is deterministic. For all pre–terminal ages, $s_t = 1$, and death occurs with certainty after period N.

2.2 Preferences

Agents maximize expected lifetime utility with constant relative risk aversion (CRRA) preferences:

$$\mathbb{E}_1 \left[\sum_{t=1}^{N} \beta^{t-1} \left(\prod_{j=1}^{t-1} s_j \right) u(c_t) \right], \qquad u(c) = \frac{c^{1-\sigma}}{1-\sigma} \tag{2}$$

where β denotes the discount factor and σ the coefficient of relative risk aversion. Following Huggett (1996), the discount factor is set to $\beta = 0.994$, rather than $\beta = 1.011$ as in the uncertain–lifetimes case, so that the certain–lifetimes model has the same average discounting once mortality risk is taken into account. The author considers two different levels of risk aversion $\sigma \in \{1.5, 3.0\}$. Lastly, labour supply is exogenous, meaning that agents do not choose hours. Instead, they receive an age– and state–dependent labour endowment.

2.3 Earnings Process

The labour endowment of an age t agent in idiosyncratic state z is:

$$e(z, t) = \exp(z + \bar{y}_t) \tag{3}$$

where $\bar{y}_t$ is the deterministic age component of log earnings and z denotes the stochastic deviation from that profile. The stochastic component evolves according to:

$$z_t = \gamma z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{4}$$

so that log labour endowment follows a process with regression toward the age–specific mean.

The initial distribution for age 1 agents is:

$$z_1 \sim \mathcal{N}(0, \sigma_{z,1}^2) \tag{5}$$

where $\sigma_{z,1}^2 = 0.38$. This parameter is calibrated jointly with the persistence parameter to match key features of the U.S. earnings distribution over the life cycle. Huggett (1996) sets $\sigma_\varepsilon^2 = 0.045$ and $\gamma = 0.96$.

The deterministic component $\exp(\bar{y}_t)$ is calibrated using data on median male earnings from the Social Security Bulletin (1981), multiplied by male labour–force participation rates from the Handbook of Labor Statistics (1985). Since Huggett (1996) reports this profile only graphically, the numerical values used in this replication are taken from the VFI Toolkit replication of Kirkby (2022), which documents data provided by Mark Huggett to Dean Corbae. The resulting profile is depicted in Figure 1.

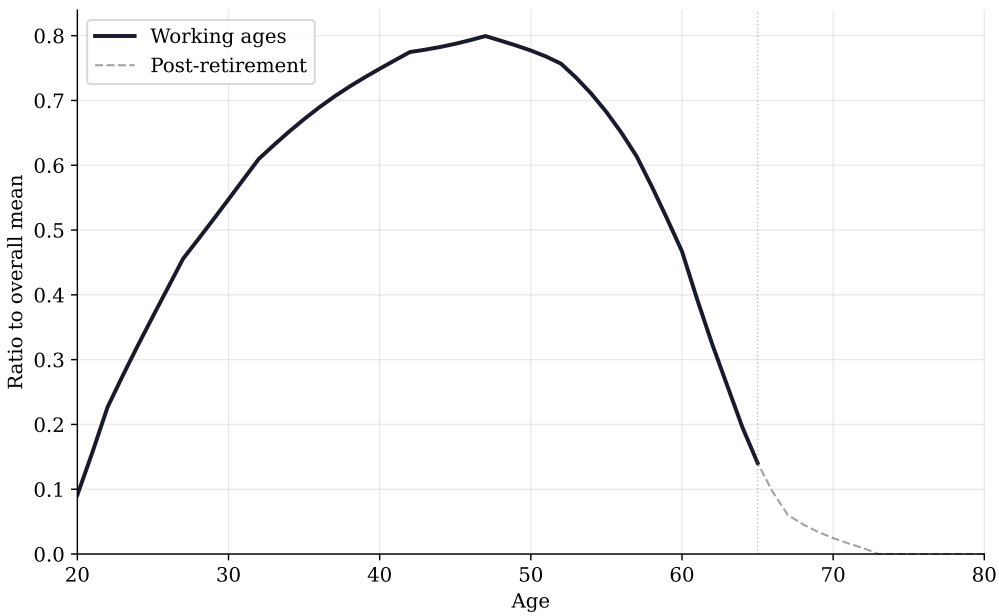


Figure 1: Age–earnings profile $\exp(\bar{y}_t)$. Numerical values from Huggett via Corbae, as used in the VFI Toolkit replication of Kirkby (2022). The profile is hump–shaped, peaking at age 47, and drops to zero by age 73.

2.4 Budget Constraint

The budget constraint for an age t agent with assets a and earnings state z is:

$$c + a' = a\big(1 + r(1-\nu)\big) + y(z,t) \tag{6}$$

where a' denotes next–period assets, r is the pre–tax interest rate, and ν is the income tax rate. The non–capital income $y(z,t)$:

$$y(z,t) = \begin{cases} (1-\theta-\nu)\, w\, e(z,t) + T & \text{if } t < R \quad \text{(working)} \\ b + T & \text{if } t \geq R \quad \text{(retired)} \end{cases} \tag{7}$$

where w is the wage, θ is the social security tax rate, b is the social security benefit, and T represents lump–sum transfers from accidental bequests. Note that both the social security tax θ and the income tax ν are applied additively to gross labour earnings, so the effective after–tax rate is $(1-\theta-\nu)$, not $(1-\theta)(1-\nu)$.

The borrowing constraint takes one of two forms: $a' \geq 0$ (no borrowing) or $a' \geq -w$ (allows agents to borrow up to one year's average earnings in the economy). In addition, agents who survive to the terminal age N must enter that period with non–negative assets. Hence, if an agent reaches age $N-1$, the savings choice must satisfy $a' \geq 0$, regardless of the borrowing limit in previous periods.

2.5 Technology and Firms

A representative firm operates a constant–returns Cobb–Douglas production function:

$$Y = AK^{\alpha}L^{1-\alpha} \tag{8}$$

where K denotes aggregate capital, L aggregate effective labour, and A total factor productivity. Under perfect competition, factor prices satisfy

$$r = \alpha A K^{\alpha-1} L^{1-\alpha} - \delta, \qquad w = (1-\alpha) A K^{\alpha} L^{-\alpha} \tag{9}$$

where δ is the depreciation rate. Huggett (1996) normalises A so that $w = 1$ when $\frac{K}{Y} = 3$ and labour input per capita equals 1.

2.6 Government

The government levies a proportional income tax at rate ν on labour income and on capital income (net of depreciation), and a payroll tax at rate θ on labour income. Government consumption satisfies:

$$G = \nu \left[wL + (r + \delta)K - \delta K \right] = \nu(wL + rK) \tag{10}$$

Following Huggett (1996), the income tax rate ν is calibrated so that $\frac{G}{Y} = 0.195$.

The social security system operates on a pay–as–you–go basis:

$$\theta wL = b \sum_{t=R}^{N} \mu_t \tag{11}$$

where b denotes the retirement benefit. Benefits are independent of an agent's past earnings history, so the system redistributes from current workers to current retirees through a flat pension financed by the payroll tax. Thus, θ lowers disposable labour income during working life and, through the balanced–budget condition, determines the level of retirement benefits.

Accidental bequests are fully taxed by the government and redistributed via lump–sum transfers to all living agents, satisfying:

$$T = \frac{1}{1+n} \sum_{t=1}^{N} \mu_t (1 - s_{t+1}) \int a'(a, z, t) \left[1 + r(1 - \nu) \right] d\psi_t(a, z) \tag{12}$$

with the convention that $s_{N+1} = 0$.

2.7 Stationary Equilibrium

A stationary equilibrium consists of policy functions $a'(a, z, t)$ and $c(a, z, t)$, prices (r, w), aggregates (K, L), transfers T, government consumption G, a social security benefit b, and a stationary distribution of agents $\{\psi_t(a, z)\}_{t=1}^{N}$ such that:

(i) **Household optimality:** given prices, taxes, transfers, and benefits, the policy functions solve the household's dynamic programming problem;

(ii) **Competitive factor pricing:** prices satisfy equation (9);

(iii) **Goods–market clearing:**

$$\sum_{t=1}^{N} \mu_t \int \left[c(a, z, t) + a'(a, z, t) \right] d\psi_t(a, z) + G = Y + (1 - \delta)K; \tag{13}$$

(iv) **Asset–market clearing:**

$$\sum_{t=1}^{N} \mu_t \int a'(a,z,t)\, d\psi_t(a,z) = (1+n)K; \tag{14}$$

(v) **Labour–market clearing:**

$$L = \sum_{t=1}^{N} \mu_t \int e(z,t)\, d\psi_t(a,z); \tag{15}$$

with retired agents contributing zero labour input through the endowment process.

(vi) **Distributional consistency:** the cross–sectional distributions evolve according to the individual policy rules and the exogenous Markov process for earnings shocks;

(vii) **Government and transfer consistency:** equations (10), (11), and (12) hold.

2.8 Calibration

Table 1 summarises the parameter values, which follow Table 2 of Huggett (1996).

Parameter	**Symbol**	**Value**	**Source / Target**
Capital share	α	0.36	NIPA
Depreciation rate	δ	0.06	NIPA
Discount factor (uncertain)	β	1.011	Hurd (1989)
Discount factor (certain)	β	0.994	Internal calibration
Risk aversion	σ	1.5 / 3.0	Table 3 / Table 4
Social security tax rate	θ	0.10	OASI
Income tax rate	ν	$\frac{\frac{G}{Y}}{1-\delta\frac{K}{Y}}$	$\frac{G}{Y} = 0.195$
AR(1) persistence	γ	0.96	Abowd and Card (1989)
Innovation variance	σ_ε^2	0.045	Abowd and Card (1989)
Initial variance	$\sigma_{y_1}^2$	0.38	PSID
Population growth	n	0.012	Census

Table 1: Calibration Parameters.

3 Computational Methods

3.1 Discretisation of the Earnings Process

For computational purposes, the continuous AR(1) process in equation (4) is approximated by a finite–state Markov chain. The implementation follows a Tauchen–style discretisation with one modification taken from Huggett (1996): the state space consists of 17 equally spaced points from $-4\sigma_{y_1}$ to $4\sigma_{y_1}$, together with a state at $6\sigma_{y_1}$ to account for extreme earnings shocks. Hence, the discretised process has 18 states in total. This construction follows the paper's strategy of assigning slightly more mass to very high earnings realisations than under a symmetric discretisation of the underlying Gaussian process.

Let $\{z_j\}_{j=1}^{18}$ denote the grid points, and let the bin boundaries be given by the midpoints between adjacent states, with the lower and upper endpoints set to $-\infty$ and $+\infty$. Conditional on current state z_i, transition probabilities are computed by integrating the conditional normal distribution $\mathcal{N}(\gamma z_i, \sigma_\varepsilon^2)$ over these bins:

$$P_{ij} = \Phi\left(\frac{b_{j+1} - \gamma z_i}{\sigma_\varepsilon}\right) - \Phi\left(\frac{b_j - \gamma z_i}{\sigma_\varepsilon}\right) \tag{16}$$

where $\Phi(\cdot)$ is the standard normal CDF and $b_j = \frac{(z_{j-1}+z_j)}{2}$ are the midpoints between adjacent grid points. The non–uniform spacing between the 17th and 18th states is automatically handled by this midpoint rule. Because the top state is placed at $6\sigma_{y_1}$, the final interval is wider than the others. Each row of the transition matrix is then renormalised to sum to one.

The initial earnings distribution for age 1 agents is constructed by integrating $\mathcal{N}(0, \sigma_{y_1}^2)$ over the same bin boundaries:

$$\pi_0(j) = \Phi\left(\frac{b_{j+1}}{\sigma_{y_1}}\right) - \Phi\left(\frac{b_j}{\sigma_{y_1}}\right) \tag{17}$$

This yields an initial distribution that is more concentrated around $z = 0$ than the stationary distribution implied by the AR(1) process. Using the stationary distribution instead would place excessive mass at extreme earnings states for newborn agents, systematically overstating initial earnings heterogeneity. Since earnings enter the budget constraint exponentially through $\exp(z)$, even moderate overdispersion at young ages can substantially increase the fraction of agents trapped at zero wealth.

In the no–shock specification, the code sets $\sigma_\varepsilon = 0$ and collapses the earnings process to a degenerate one–state chain with $z = 0$ and transition probability equal to one. In that case, the initial distribution is also degenerate.

3.2 Asset Grid

Asset holdings are discretised on a uniform grid:

$$\mathcal{A} = \{a_1, a_2, \ldots, a_{n_k}\} \tag{18}$$

constructed as an evenly spaced grid between a lower bound $a_{\min}$ and an upper bound $a_{\max}$. The lower bound is set to $a_{\min} = 0$ under the no–borrowing specification and to $a_{\min} = -w$ when agents are allowed to borrow up to one year's wage income. The upper bound $a_{\max}$ and the number of grid points n_k are fixed ex ante. In the implementation, the exact grid spacing is therefore implied by the chosen values of $a_{\min}$, $a_{\max}$, and n_k.

This choice follows Huggett (1996), who uses uniform asset grids and allows the number of grid points to vary across model economies. I retain a uniform 301 point grid to remain close to the paper's numerical design, taking advantage of modern computational power, while keeping the EGM implementation transparent.

3.3 Solving the Household Problem: Endogenous Grid Method

The household's problem is solved by backward induction from the terminal age using the Endogenous Grid Method (EGM) of Carroll (2006). The EGM exploits the Euler equation to avoid the costly non–linear maximisation required by standard value function iteration (VFI), which Huggett (1996) originally employed.

At an interior solution, the household's optimality condition (Euler equation) is:

$$u'(c_t) = \beta\, s_{t+1}\, R\, \mathbb{E}\big[u'(c_{t+1})\big] \tag{19}$$

where $R = 1 + r(1 - \nu)$ is the gross after–tax return and the expectation is taken over tomorrow's earnings state using the transition matrix P.

At the terminal age N, the agent consumes all available resources: $c_N = R \cdot a + b + T$ and $a' = 0$. At age $N - 1$, the constraint $a' \geq 0$ is imposed regardless of the economy's borrowing limit, since assets carried into the terminal period must be non–negative.

EGM step: For each earlier age t, working backward:

1. Compute the expected marginal utility of consumption tomorrow for each pair of current earnings state z_i and possible savings choice a'_k:

$$\overline{u}'(z_i, a'_k) = \sum_{j=1}^{n_z} P_{ij}\, u'\big(c_{t+1}(z_j, a'_k)\big) \tag{20}$$

2. Invert the Euler equation to obtain the consumption that is optimal *given* the choice a'_k:

$$c^*(z_i, a'_k) = \big(\beta\, s_{t+1}\, R\, \overline{u}'(z_i, a'_k)\big)^{-\frac{1}{\sigma}} \tag{21}$$

3. Use the budget constraint to find the *implied* current assets (the "endogenous grid point"):

$$a^{\text{implied}}(z_i, a'_k) = \frac{c^*(z_i, a'_k) + a'_k - y(z_i, t)}{R} \tag{22}$$

4. The pairs $\{a_k^{\text{implied}}, a'_k\}$ define the policy function on an endogenous grid. Interpolate back onto the exogenous asset grid $\mathcal{A}$ using linear interpolation to obtain $a'(a, z_i, t)$ for each $a \in \mathcal{A}$.

5. For agents whose assets fall below the lowest endogenous grid point, the borrowing constraint binds: $a' = a_{\min}$ and $c = R \cdot a + y - a_{\min}$.

The key advantage of EGM over VFI is that its step 2 provides a closed–form solution for consumption, eliminating the need for non–linear root finding or grid search at each state. As seen in Goensch (2025), EGM achieves speedups of approximately $30\times$ over VFI with golden section search for OLG models of comparable complexity. Both methods should yield equivalent solutions. The choice of EGM is purely out of computational convenience.

3.4 Computing the Wealth Distribution

Given the policy functions, the stationary wealth distribution must be computed to evaluate the market clearing conditions. Huggett (1996) computes the wealth distribution using distributional iteration, as described in Huggett (1993). This method tracks the probability distribution $\psi_t(a, z)$ directly on the grid, iterating forward age by age using the policy function and transition matrix. When the policy function maps a grid point to an asset level a' that falls between grid points, the probability mass is split between the two nearest grid points in proportion to distance (the so called "lottery" method). Distributional iteration is exact on the grid and introduces no sampling noise.

Instead, I simulate a panel of $N_{\text{sim}} = 500{,}000$ agents through their entire life cycle. Each agent begins at age 20 with zero assets and an earnings state drawn from π_0 using the inverse CDF method, as described earlier. At each age, the agent's optimal savings is obtained by linearly interpolating the policy function at their actual wealth level. Next–period earnings are drawn using the same inverse CDF method applied to the relevant row of the cumulative transition matrix.

The advantage of Monte Carlo simulation is its simplicity. It avoids the probability–splitting logic and does not restrict agents' asset holdings to grid points. The clear disadvantage is sampling noise. With 500,000 agents, the top 1% comprises only 5,000 individuals, so top wealth shares exhibit some variability across different random seeds. For the aggregate moments reported in the paper ($\frac{K}{Y}$, Gini, top shares to one decimal place), the noise is small relative to the precision required.

3.5 General Equilibrium Algorithm

The stationary equilibrium is found by iterating on two aggregate variables: capital K and lump–sum transfers T.

Algorithm 1 General Equilibrium Iteration

1: **Initialise:** Set $K^{(0)}$, $T^{(0)}$, choose tolerances ε_K, ε_T.
2: **for** $s = 1, 2, \ldots$ **do**
3: **Prices:** Compute r, w, R, b from $K^{(s-1)}$ using (9)–(11).
4: **Household problem:** Solve backward via EGM to obtain $a'(a, z, t)$.
5: **Distribution:** Simulate 500,000 agents forward to obtain current assets $a_{i,t}$ and implied next–period assets $a'_{i,t}$.
6: **Aggregates:** Compute model–implied capital and transfers:

$$K^* = \frac{1}{1+n} \sum_{t=1}^{N} \mu_t \, \bar{a}'_t, \qquad T^* = \frac{1}{1+n} \sum_{t=1}^{N} \mu_t (1 - s_{t+1}) \, R \, \bar{a}'_t.$$

7: **Convergence:** If $|K^* - K^{(s-1)}| < \varepsilon_K$ and $|T^* - T^{(s-1)}| < \varepsilon_T$, stop.
8: **Update with dampening:**

$$K^{(s)} = (1 - \lambda_K) K^{(s-1)} + \lambda_K K^*, \qquad T^{(s)} = (1 - \lambda_T) T^{(s-1)} + \lambda_T T^*.$$

9: **end for**

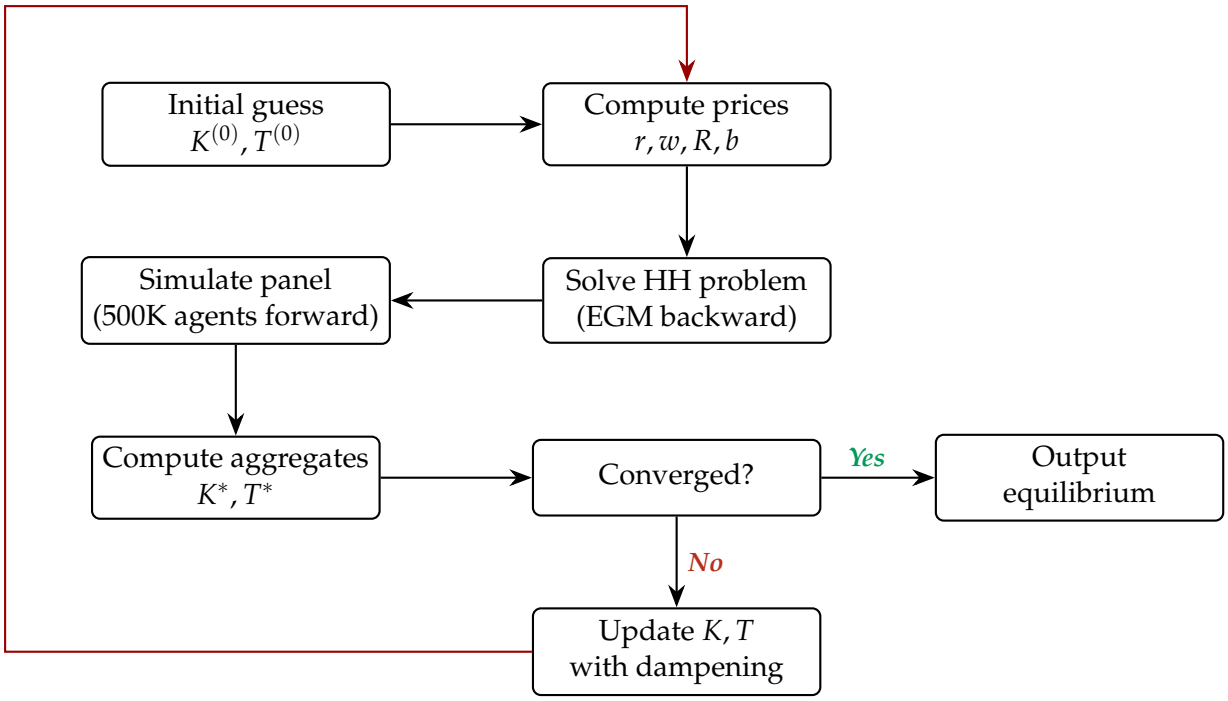


Figure 2: General equilibrium algorithm. The red arrow indicates the iteration loop.

3.6 Summary of Computational Choices

Table 2 summarises the computational choices made in this replication compared with the original paper.

Component	This replication	Huggett (1996)
Household problem	EGM (Carroll, 2006)	Value function iteration
Asset grid	Uniform, 301 points	41–301 points depending on economy
Earnings discretisation	18–state Tauchen variant	18–state discretisation
Wealth distribution	Monte Carlo (500K agents)	Distributional iteration
Earnings profile	Kirkby (2022)	SSA Bulletin / Handbook of LS
GE updating	Damped fixed–point on K and T	Not specified

Table 2: Computational Methods: This Replication vs. Original Paper

The code is implemented in Python and organised into six modules:

1. `earnings.py` (Markov chain discretisation);
2. `household.py` (preferences, technology, EGM);
3. `simulation.py` (Monte Carlo panel simulation);

4. `statistics.py` (distributional moments);

5. `equilibrium.py` (General equilibrium solver);

6. `calibration.py` (parameters and data).

The EGM and simulation functions are compiled using Numba `@njit` for performance (see `README.md` for further software and hardware details).

4 Results

4.1 Baseline Results

Table 3 reports the baseline moments for the paper's benchmark specification: uncertain lifetimes, idiosyncratic earnings shocks ($\sigma_\varepsilon^2 = 0.045$), borrowing limit $a' \geq -w$, and $\sigma = 1.5$. Although the author does not explicitly define this case as the baseline economy, it is the paper's primary reference. U.S. data are taken from Huggett (1996).

Moment	**Replication**	**Paper**	**U.S. Data**
$\frac{K}{Y}$	3.323	3.2	3.0
Gini coefficient	0.746	0.76	0.72
Top 1% share	11.8%	11.8%	28.0%
Top 5% share	35.4%	35.6%	53.5%
Top 20% share	74.6%	75.5%	79.5%
Fraction ≤ 0	28.9%	24.0%	15.0–20.0%

Table 3: Baseline Results ($\sigma = 1.5$, uncertain lifetimes, shocks, $a' \geq -w$)

The baseline fit is strong. The capital–output ratio, wealth Gini, and top wealth shares are all close to the paper's values, with the top 1% share matching exactly at one decimal place. The main discrepancy is the fraction of households with zero or negative wealth (28.9% vs 24.0%). This discrepancy most likely reflects Monte Carlo sampling noise, differences in the discretisation of the earnings process, and the use of the Corbae/Huggett age–earnings profile.

Figure 3 is broadly consistent with Figure 2 of Huggett (1996) and with the corresponding figure in the VFI Toolkit replication of Kirkby (2022) (see Figure 9 in Appendix for a visual comparison). In all cases, both mean and median wealth are hump–shaped over the life cycle: they rise during the working years, peak shortly before retirement, and decline thereafter, reflecting life–cycle accumulation before retirement and decumulation during retirement. Wealth dispersion increases through middle age and narrows at older ages, as shown by the evolution of the $p10$–$p90$ and $p25$–$p75$ ranges. The

gap between mean and median wealth reflects the positive skewness of the wealth distribution. Figure 4 exhibits similar patterns. Specifications with earnings shocks generate higher peak wealth due to the precautionary saving motive. As depicted in Figure 5, shocks accentuate inequality, leading to substantially greater wealth concentration (Lorenz curve bows further from the 45° line).

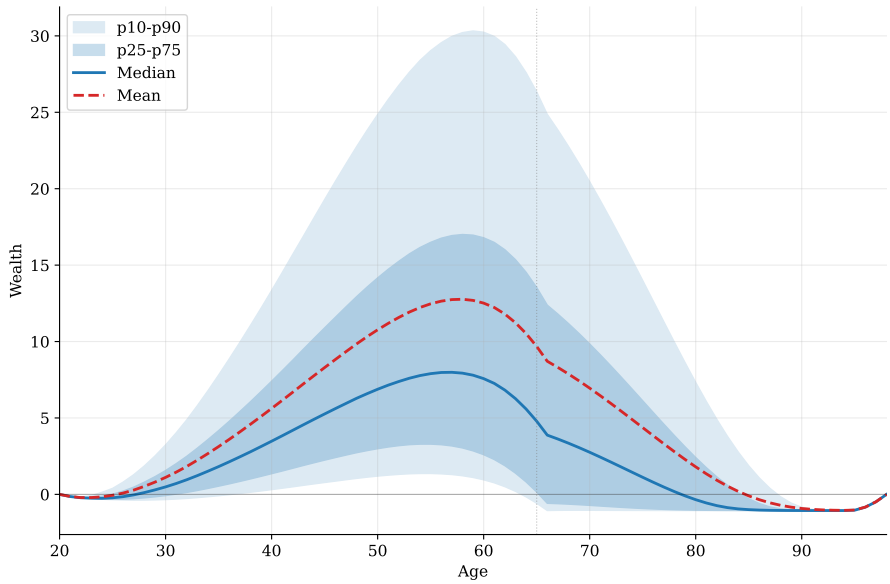


Figure 3: Wealth percentile bands by age for the baseline specification (uncertain lifetimes, shocks, $a' \geq -w$).

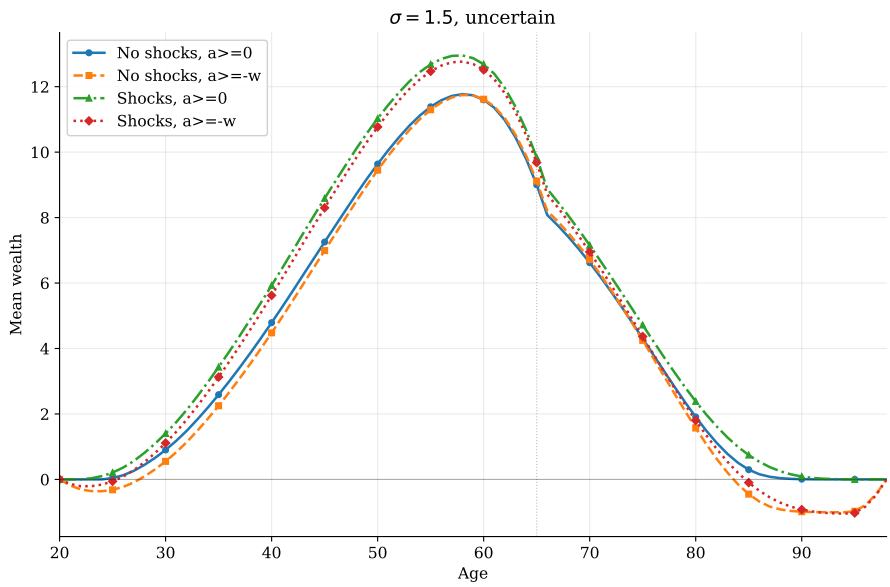


Figure 4: Mean wealth by age for the four uncertain–lifetimes specifications ($\sigma = 1.5$).

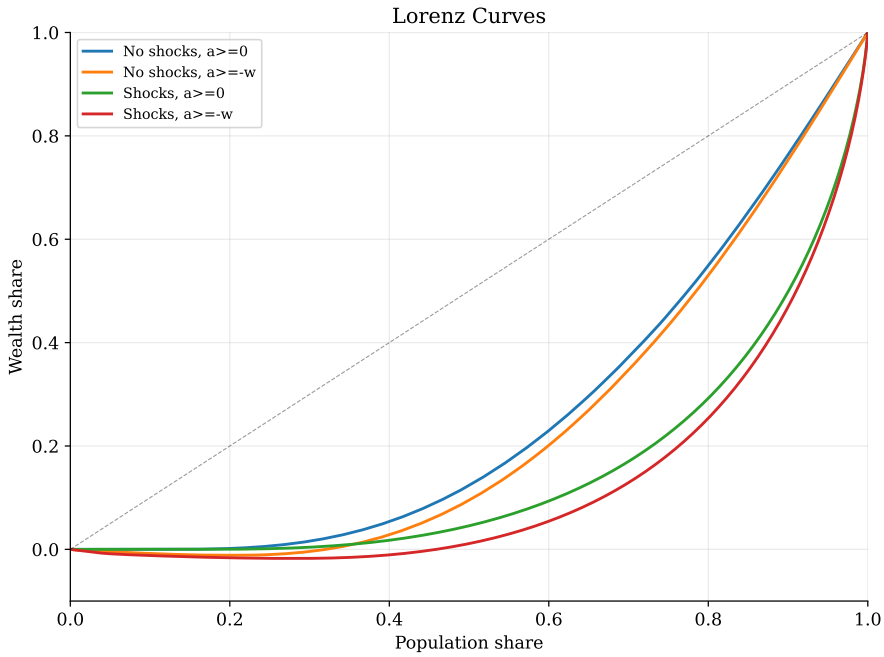


Figure 5: Lorenz curves for the four uncertain–lifetimes specifications ($\sigma = 1.5$).

4.2 Replication of Table 3

Table 4 reports results for all eight specifications with $\sigma = 1.5$, alongside the corresponding values reported in the paper. Computing these results, together with the replication of Huggett's Table 4, was the most time–consuming part of the exercise, since it required solving 8 model economies, each with 500,000 simulated agents, 18 earnings states, 79 ages and 301 asset grid points.

		$\frac{K}{Y}$		**Gini**		**Frac** ≤ 0	
Lifetimes	**Specification**	Repl.	Paper	Repl.	Paper	Repl.	Paper
Certain	No shocks, $a' \geq 0$	4.64	2.9	0.463	0.47	14.9%	14.0%
Certain	No shocks, $a' \geq -w$	4.47	2.8	0.507	0.54	21.9%	25.0%
Certain	Shocks, $a' \geq 0$	5.00	3.2	0.662	0.70	14.5%	19.0%
Certain	Shocks, $a' \geq -w$	4.85	3.1	0.711	0.74	27.6%	24.0%
Uncertain	No shocks, $a' \geq 0$	3.12	3.1	0.482	0.46	12.9%	11.0%
Uncertain	No shocks, $a' \geq -w$	3.03	3.0	0.518	0.49	21.3%	12.0%
Uncertain	Shocks, $a' \geq 0$	3.43	3.4	0.692	0.69	12.8%	17.0%
Uncertain	Shocks, $a' \geq -w$	3.32	3.2	0.746	0.76	28.9%	24.0%

Table 4: Replication of Huggett's Table 3 ($\sigma = 1.5$)

The uncertain–lifetimes specifications (lower panel) match the paper closely in the capital–output ratio and wealth concentration measures. In particular, the baseline economy with shocks and borrowing has $\frac{K}{Y} = 3.32$ against 3.2 in the paper, and

top wealth shares that are nearly identical. The largest discrepancies are again in the fraction of households with zero or negative wealth, especially in the borrowing economies. The certain–lifetimes specifications (upper panel) show a systematic upward bias in $\frac{K}{Y}$ (roughly 4.5–5.0 versus the paper's 2.8–3.2), indicating that perhaps the certain–lifetimes calibration interacts strongly with the age–earnings profile used here.

Below, Figure 6 displays the wealth profiles for the baseline replication. The figure mimics Figure 3 of Huggett (1996), which reports data for the U.S. economy (see Figure 12 in the Appendix). Once again, in both figures, mean wealth, median wealth, and the quantiles display pronounced hump–shaped life–cycle profiles, rising during the working years, peaking around ages 50–60, and declining thereafter. The replication captures well the ordering of the moments across the distribution, their overall behaviour, and the location and timing of their peaks. The fit is especially strong for the mean and median profiles. The main discrepancies arise in the lower tail. Relative to Huggett's figure, the 25th and 10th quantiles in the replication decline somewhat more sharply after retirement and become negative earlier.

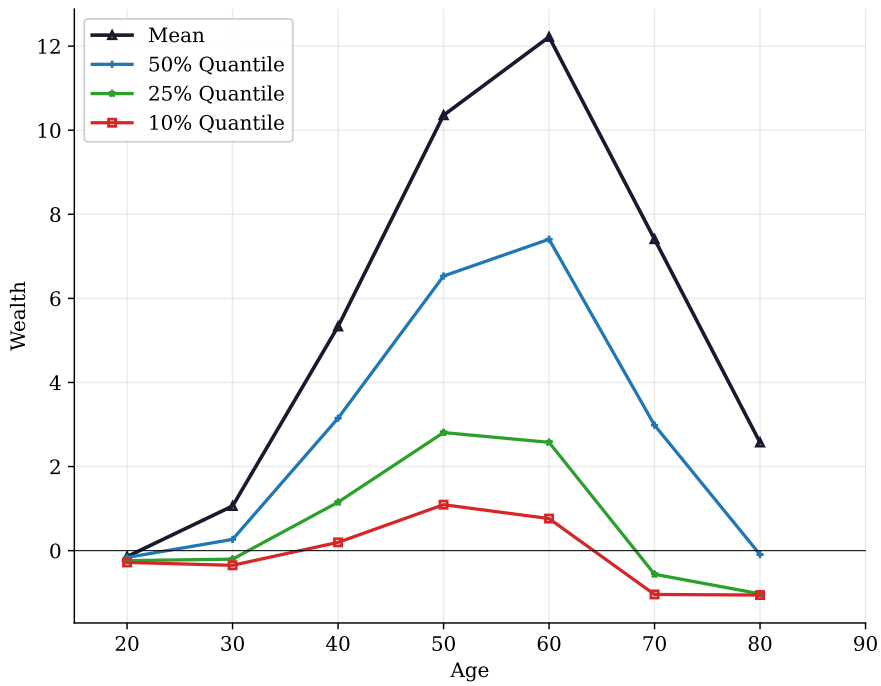


Figure 6: Replication of Huggett's Figure 3: wealth profiles for the baseline specification (uncertain lifetimes, shocks, $a' \geq -w$, $\sigma = 1.5$). Statistics are computed within 10–year age bins.

4.3 Replication of Table 4

Table 5 reports results for all eight specifications with $\sigma = 3.0$, alongside the corresponding values in the paper. As with Table 4 (case where $\sigma = 1.5$), computing these results was computationally intensive, since it required solving the same set of eight model economies under the alternative preference parameterisation.

		$\frac{K}{Y}$		**Gini**		**Frac** ≤ 0	
Lifetimes	**Specification**	Repl.	Paper	Repl.	Paper	Repl.	Paper
Certain	No shocks, $a' \geq 0$	4.34	2.3	0.472	0.51	14.9%	21.0%
Certain	No shocks, $a' \geq -w$	4.07	2.0	0.527	0.62	25.2%	29.0%
Certain	Shocks, $a' \geq 0$	5.09	2.9	0.652	0.66	11.8%	3.0%
Certain	Shocks, $a' \geq -w$	4.92	2.8	0.693	0.73	24.7%	23.0%
Uncertain	No shocks, $a' \geq 0$	2.45	2.5	0.527	0.50	20.3%	21.0%
Uncertain	No shocks, $a' \geq -w$	2.25	2.3	0.627	0.61	34.2%	29.0%
Uncertain	Shocks, $a' \geq 0$	3.14	3.0	0.708	0.72	14.4%	19.0%
Uncertain	Shocks, $a' \geq -w$	2.99	2.8	0.778	0.84	33.1%	40.0%

Table 5: Replication of Huggett's Table 4 ($\sigma = 3.0$)

Similarly to the case with $\sigma = 1.5$, the uncertain–lifetimes results are reasonably close in terms of $\frac{K}{Y}$ and the wealth concentration statistics, whereas the certain–lifetimes specifications continue to produce capital–output ratios well above the values reported in the paper. The stronger risk–aversion case also preserves the qualitative pattern that earnings risk and looser borrowing constraints increase wealth concentration.

4.4 Replication of Figure 4 & 5

Figures 7 and 8 reproduce the main patterns reported in Figures 4 and 5 of Huggett (1996) and are also closely aligned with the corresponding plots in the VFI Toolkit replication of Kirkby (2022) (see Figures 10 and 11 in the Appendix). In both the certain– and uncertain–lifetimes economies, within–cohort wealth inequality is higher under the looser borrowing constraint ($a' \geq -w$) than under the no–borrowing specification ($a' \geq 0$), particularly at younger ages. In the certain lifetimes, the gap narrows over the life cycle, and the Gini profile declines from young adulthood into middle age before flattening and rising only modestly near retirement. Regarding the uncertain lifetimes, the profile is more strongly U–shaped. Inequality falls through middle age but rises sharply at older ages, especially under the looser borrowing constraint. This late–life increase is consistent with the cumulative effect of persistent idiosyncratic shocks among agents who survive to older ages.

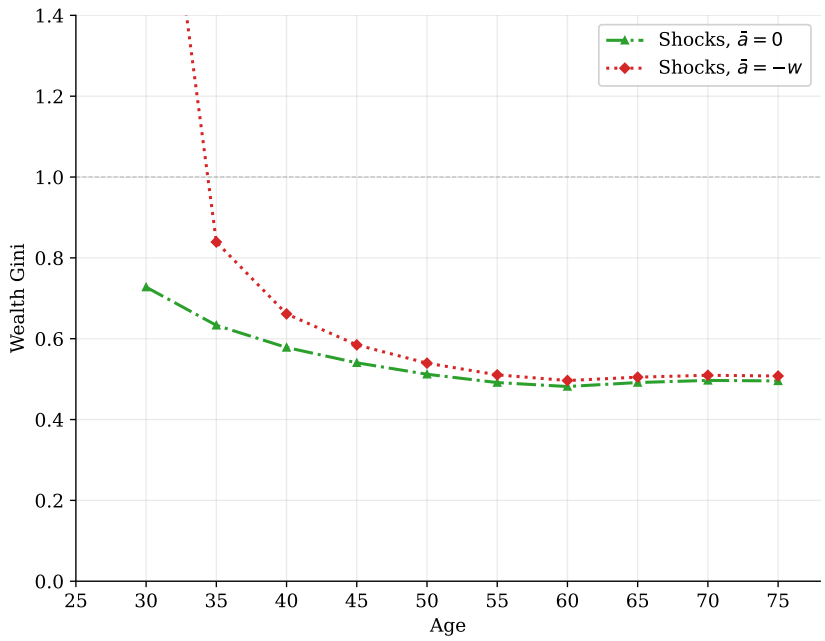


Figure 7: Replication of Figure 4: Gini coefficients within 5–year age groups, certain lifetimes, with earnings shocks ($\sigma = 1.5$).

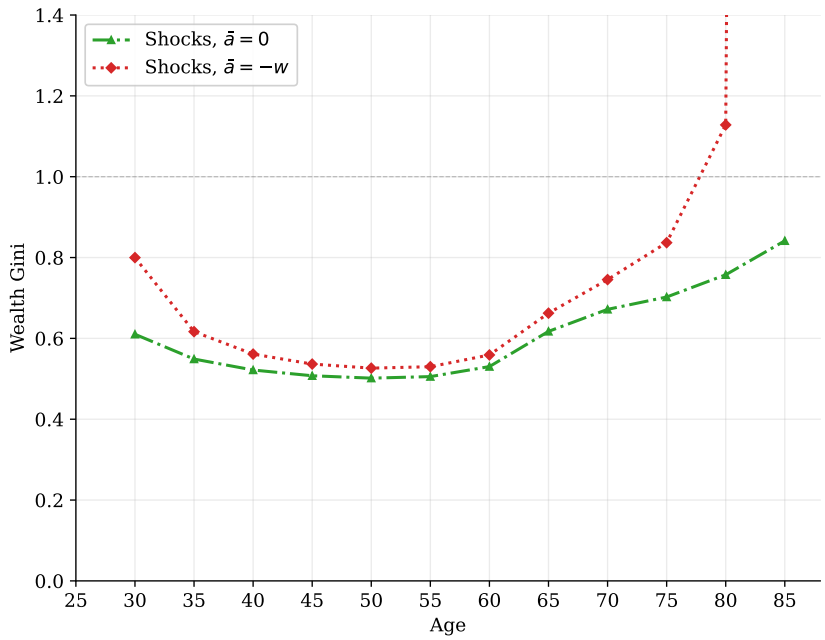


Figure 8: Replication of Figure 5: Gini coefficients within 5–year age groups, uncertain lifetimes, with earnings shocks ($\sigma = 1.5$).

4.5 Discussion of Results

The replication reproduces the main economic patterns documented in Huggett (1996). Introducing earnings shocks substantially increases wealth inequality: in the uncertain–lifetimes economy with $a' \geq -w$, the Gini rises from 0.518 to 0.746 when shocks are switched on, as agents who receive persistently high (low) earnings realisations accumulate substantially more (less) wealth. Relaxing the borrowing constraint from $a' \geq 0$ to $a' \geq -w$ raises both the Gini and the fraction of households with zero or negative wealth across all specifications. Higher risk aversion amplifies concentration in the uncertain–lifetimes economies, with the $\sigma = 3.0$ specifications generally exhibiting higher Gini coefficients and larger top wealth shares than their $\sigma = 1.5$ counterparts.

The main discrepancy is that the certain–lifetimes specifications generate $\frac{K}{Y}$ ratios systematically above the values reported in the paper, while the uncertain–lifetimes specifications are much closer to the benchmark moments. A possible explanation has to do with the Corbae earnings profile, which drops to exactly zero at age 73, creating 25 years of zero labour income for agents who survive to 98 with certainty. Under uncertain lifetimes, most agents die before this region becomes quantitatively important, so its effect on aggregate saving is muted. By contrast, under certain lifetimes, every agent must finance a long deterministic retirement spell with no labour income, creating a strong saving motive during working life that perhaps explains the upward bias in $\frac{K}{Y}$.

The earnings process also helps explain why wealth inequality is much larger in the shock economies. At age 20, the cross–sectional variance of log earnings equals $\sigma^2_{y_1} = 0.38$. As agents age, idiosyncratic shocks accumulate and the variance rises monotonically toward the unconditional AR(1) variance $\sigma^2_\varepsilon/(1-\gamma^2) \approx 0.574$. This widening dispersion in labour income translates directly into greater heterogeneity in savings behaviour, generating the stronger wealth concentration observed in the shock economies.

5 Conclusion

This paper provides a computational replication of Huggett (1996). The replication matches the paper's uncertain–lifetimes baseline closely: $\frac{K}{Y} = 3.323$ (paper: 3.2), Gini $= 0.746$ (paper: 0.76), top 1% wealth share $= 11.8\%$ (paper: 11.8%), and top 5% wealth share $= 35.4\%$ (paper: 35.6%). The main qualitative patterns – namely the effects of earnings shocks, borrowing constraints, and risk aversion on wealth concentration – are robustly reproduced across all specifications.

The replication departs from the original paper in two computational respects. The household problem is solved using the Endogenous Grid Method (Carroll, 2006) rather than value function iteration, and the stationary wealth distribution is computed via Monte Carlo simulation of 500,000 agents rather than distributional iteration (Huggett, 1993). These choices simplify the implementation at the cost of introducing some sampling noise.

The main discrepancy is that the certain–lifetimes economies generate $\frac{K}{Y}$ ratios well above the paper's values, as discussed in Section 4.5. Several extensions could further improve the replication: implementing distributional iteration to eliminate sampling noise, recalibrating β for the certain–lifetimes case under the Corbae earnings profile, and exploring denser asset grids near the borrowing constraint to improve measurement of the lower tail of the wealth distribution.

6 Appendix

Appendix Figures 9–12 report the corresponding figures from Huggett (1996) and the VFI Toolkit replication of Kirkby (2022) for a visual result comparison.

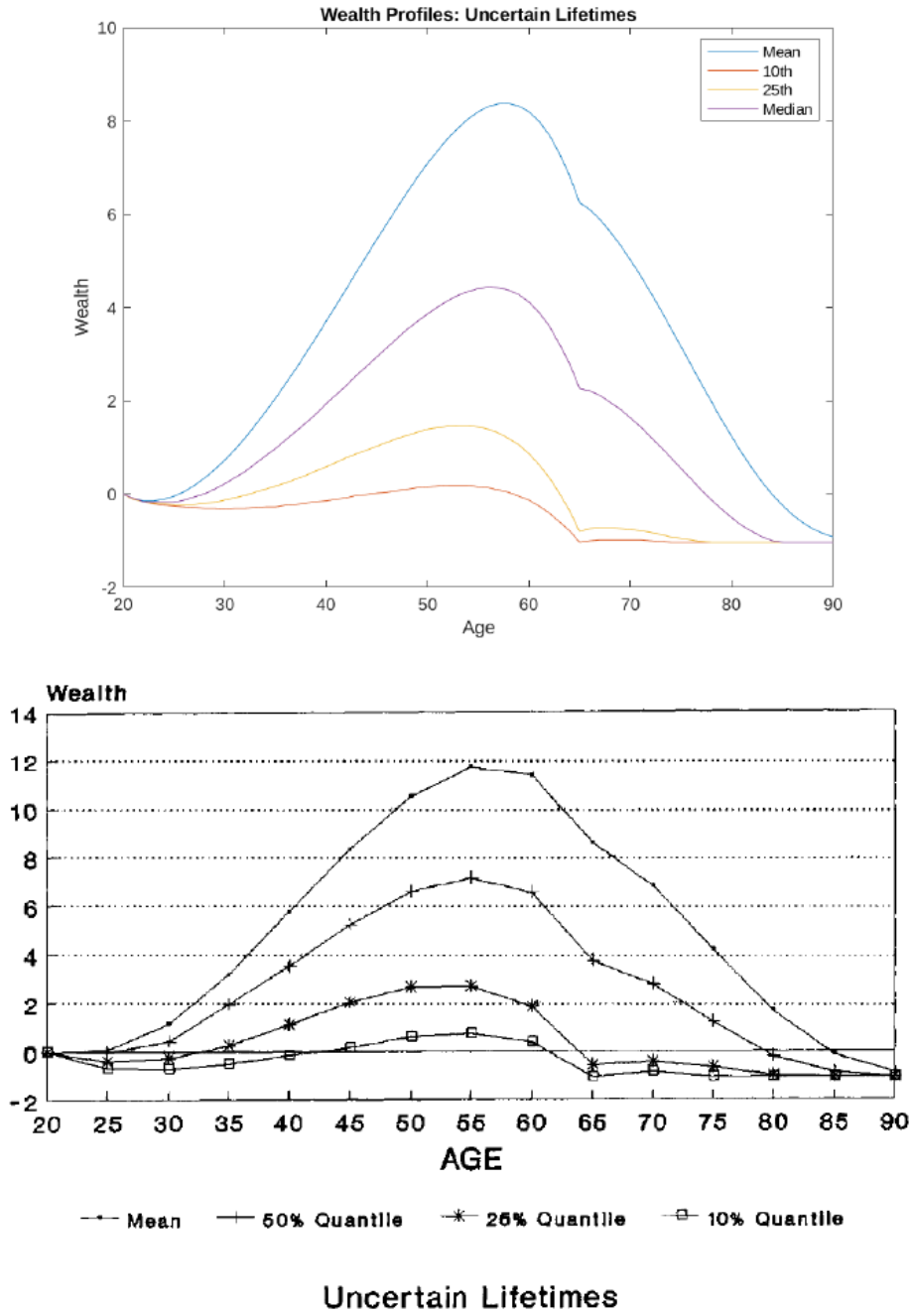
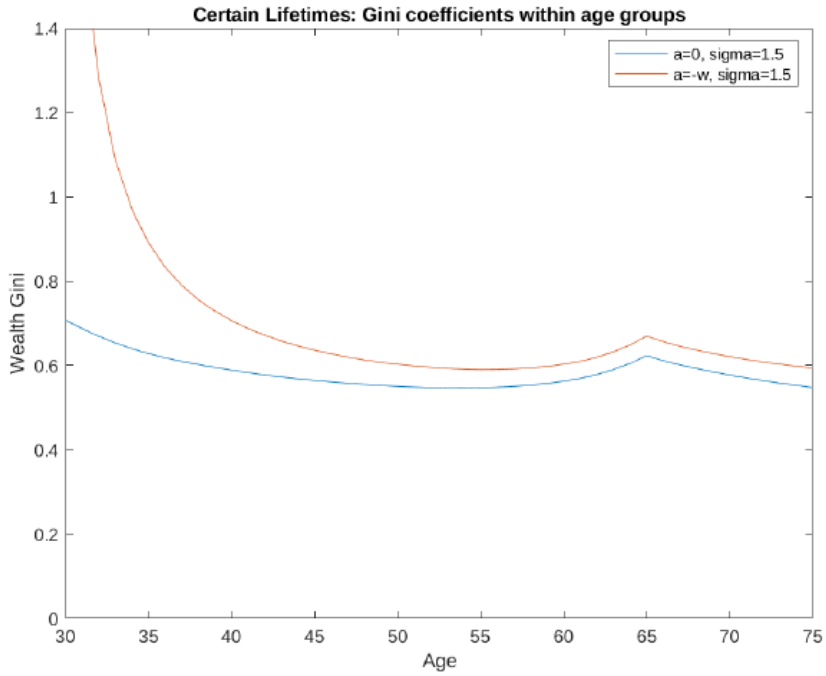


Figure 9: Reference figures for wealth profiles in the uncertain–lifetimes economy. The upper panel reports the corresponding figure from the VFI Toolkit replication of Kirkby (2022), while the lower panel shows Figure 2 of Huggett (1996).



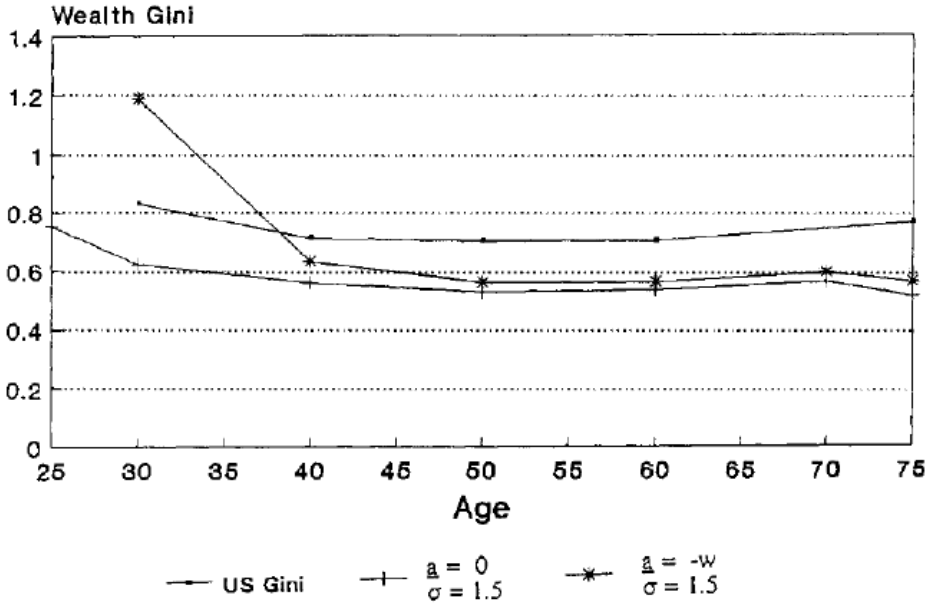


Figure 10: Reference figures for within–age wealth Gini coefficients in the certain–lifetimes economy. The upper panel reports the corresponding figure from the VFI Toolkit replication of Kirkby (2022), while the lower panel shows Figure 4 of Huggett (1996).

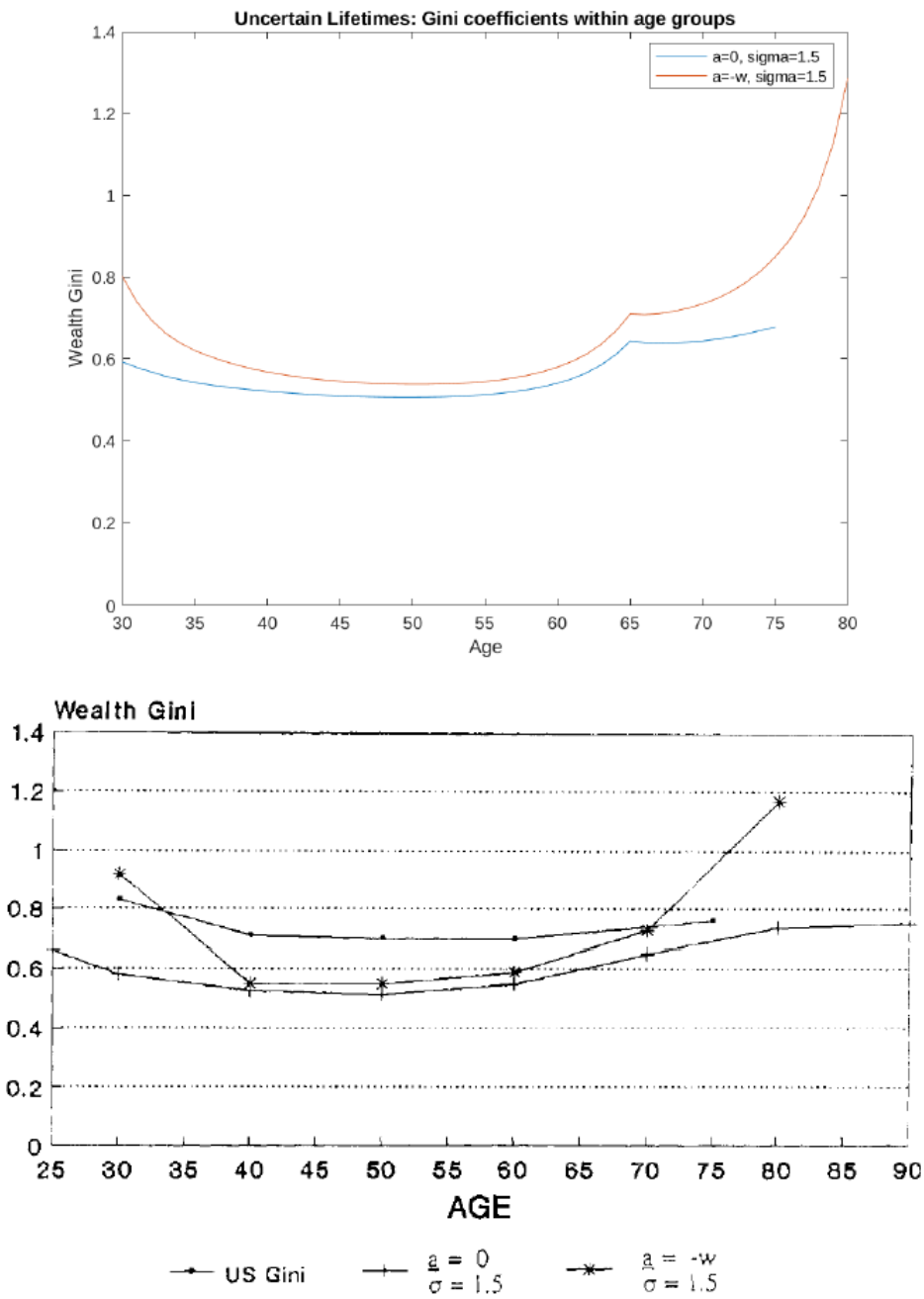


Figure 11: Reference figures for within–age wealth Gini coefficients in the uncertain–lifetimes economy. The upper panel reports the corresponding figure from the VFI Toolkit replication of Kirkby (2022), while the lower panel shows Figure 5 of Huggett (1996).

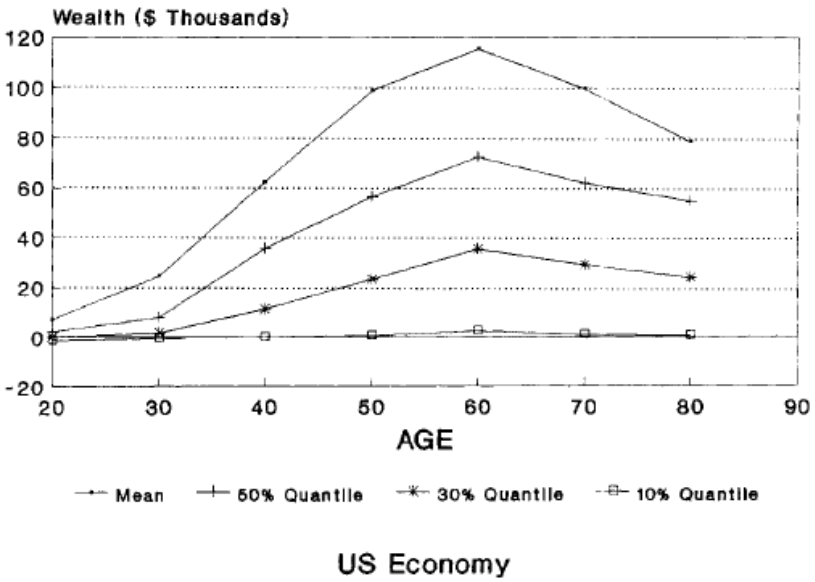


Fig. 3. Wealth profiles.

Figure 12: Reference figure for wealth profiles in U.S. Economy. Figure 3 of Huggett (1996).

References

Abowd, J. M. and Card, D. (1989). On the covariance structure of earnings and hours changes. *Econometrica*, 57(2):411–445.

Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.

Goensch, J. (2025). Lecture 6: OLG — Uncertainty. Quantitative Macroeconomics and Numerical Methods, Goethe University Frankfurt.

Huggett, M. (1993). The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control*, 17(5–6):953–969.

Huggett, M. (1996). Wealth distribution in life-cycle economies. *Journal of Monetary Economics*, 38(3):469–494.

Hurd, M. D. (1989). Mortality risk and bequests. *Econometrica*, 57(4):779–813.

Jordan, C. W. (1975). *Life Contingencies*. Society of Actuaries.

Kirkby, R. (2022). VFI Toolkit: Replication of Huggett (1996). GitHub repository. Includes age-earnings profile data from Mark Huggett via Dean Corbae.